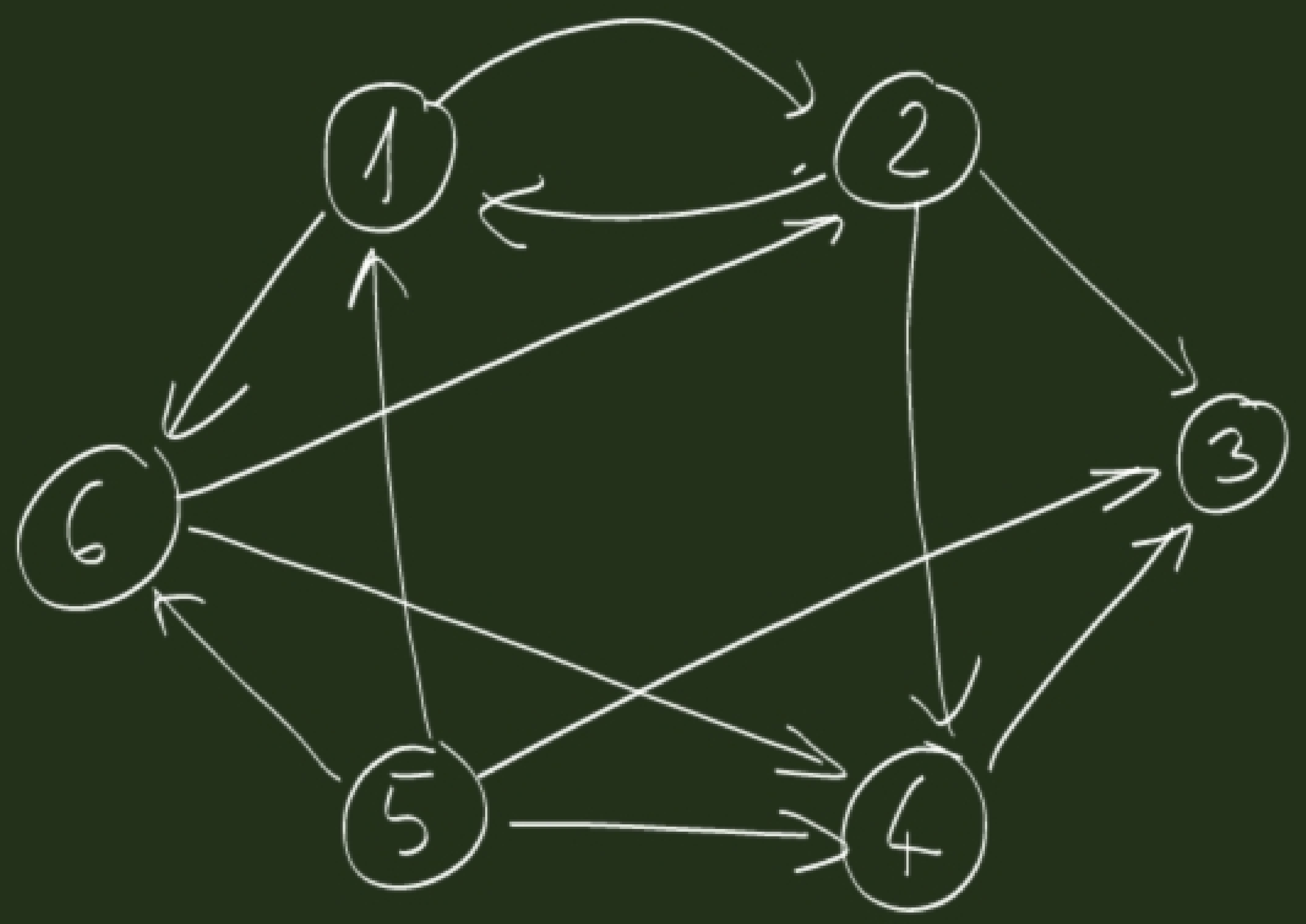
graph G with $G.V = \{1, 2, 3, \dots, n\}$

Transitive closure of G is $T \in \mathbb{B}[n, n]$

such that

$$\underset{\text{true}}{T[i, j] = 1} \iff \exists \text{ path } i \rightsquigarrow j$$



$\exists\ i \rightsquigarrow i$

$2 \rightsquigarrow$

$$T = \begin{array}{c|cccccc|} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 1 & 1 & 0 & 1 \\ 3 & 0 & 0 & 1 & 0 & 0 & 0 \\ 4 & 0 & 0 & 1 & 1 & 0 & 0 \\ 5 & 1 & 1 & 1 & 1 & 1 & 1 \\ 6 & 1 & 1 & 1 & 1 & 0 & 1 \\ \hline \end{array}$$

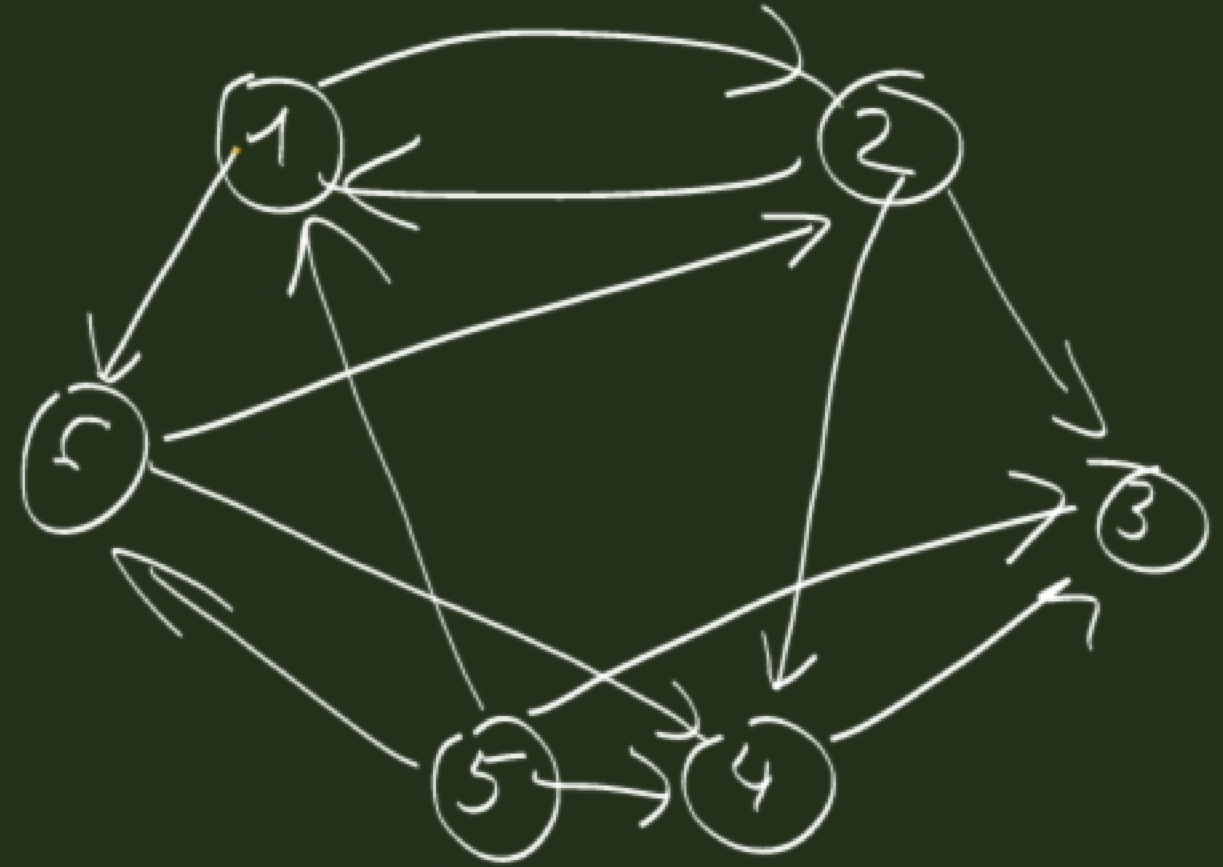
1st method : perform BFS from each vertex

$\cancel{\text{BFS}}(G:\mathcal{G}\,;\,s:\mathcal{V})$ BFS_TC

- $\forall u \in G.V$
  - $\cancel{d(u) := \infty\,;\,\pi(u) := \emptyset}$
  - $color(u) := white$
- $\cancel{d(s) := 0}$; $color(s) := grey$
- $Q : Queue\,;\,Q.\text{add}(s)$
- $\neg Q.\text{isEmpty}()$
  - $u := Q.\text{rem}()$
  - $\forall v \in G.A(u)$
    - $\cancel{d(v) = \infty}$ color(v)=white
      - $\cancel{d(v) := d(u)+1}$
      - $\cancel{\pi(v) := u}$
      - $color(v) := grey$
      - $Q.\text{add}(v)$
    - else: SKIP
  - $\cancel{[color(u) := black]}$

TransitiveClosure$(G:\mathcal{G}) : \mathbb{B}[n,n]$

- $i = 1$ to n
  - $BFS\_TC(G, i)$
  - $j = 1$ to n
    - color(j)=grey
      - $T[i,j] := 1$
    - else: $T[i,j] := 0$



$1 \to 2\,;\,6.$
$2 \to 1\,;\,3\,;\,4.$
$3 \to .$
$4 \to 3.$
$5 \to 1\,;\,3\,;\,4\,;\,6.$
$6 \to 2\,;\,4.$

source	colors of vertices 1	2	3	4	5	6	Q	vert. expanded
s=1	w	w	w	w	w	w	< >	–
	g	w	w	w	w	w	<1>	–
	g	g	w	w	w	g	<2,6>	1
	g	g	g	g	w	g	<6,3,4>	2
							<3,4>	6
							<4>	3
							< >	4

$T =$

1	1	1	1	0	1

$1 \longrightarrow 2;6.$

$2 \longrightarrow 1;3;4.$

$3 \longrightarrow .$

$4 \longrightarrow 3$

$5 \longrightarrow 1;3;4;6.$

$6 \longrightarrow 2;4.$

	1	2	3	4	5	6	Q	v.exp
$\boxed{s=2}$	w	g	w	w	w	w	$\langle 2 \rangle$	—
	g	g	g	g	w	w	$\langle 1,3,4 \rangle$	2
						g	$\langle 3,4,6 \rangle$	1
							$\langle 4,6 \rangle$	3
							$\langle 6 \rangle$	4
							$\langle \ \rangle$	6
	g	g	g	g	w	g		

$$T = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ & & & & & \end{pmatrix}$$

and so on...

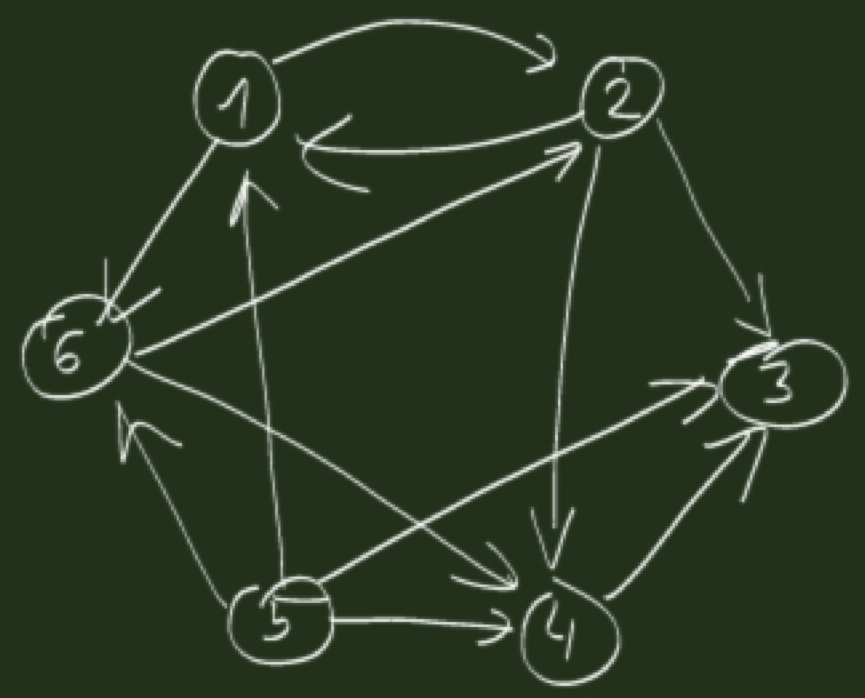
2nd method

```
TransitiveClosure(A/1, T/1 : 𝔹[n, n])
  i := 1 to n
    j := 1 to n
      T[i, j] := A[i, j]
    T[i, i] := 1
  k := 1 to n
    i := 1 to n
      j := 1 to n
        T[i, j] := T[i, j] ∨ (T[i, k] ∧ T[k, j])
```

$$T = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & \cancel{0} & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

1

1



adj. mtx:

$$A = \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

$\boxed{k = 1}$

$$\begin{array}{c|c} & \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & & \\ 0 & 0 & 1 & = & & \\ 0 & & & & & \\ 1 & & 1 & 1 & & \\ 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

$T[3,4] =$

$= \text{false} \lor$

$\lor \underbrace{[\underbrace{T[3,1]}_{F} \land \underbrace{T[1,4]}_{F}]}_{F} = F$

```
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      T[i, j] := A[i, j]
    T[i, i] := 1
  k := 1 to n
    i := 1 to n
      j := 1 to n
        T[i, j] := T[i, j] ∨ (T[i, k] ∧ T[k, j])
```

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$T^{(0)} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

$k = 1$

$$T^{(1)} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

$k = 2$

$$T^{(2)} = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$T^{(3)} = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \end{bmatrix} = T^{(2)}$$

$$T^{(4)} = T^{(3)}$$

$$T^{(5)} = T^{(4)} = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$T^{(6)} = T^{(5)}$$

Same matrix...